ARTICLE 15 October 2026

How autonomous research shapes scientific understanding

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Temporary ImageGen figure preview. Generated geometry is illustrative; quantitative evidence is in the text and tables.

Autonomous science promises to translate experimental agency into advances in scientific understanding[1,2]. Here we use ChemWorld, a set of programmable chemical worlds, to examine how self-directed research forms relationships that guide subsequent judgments. Ninety repeated process sessions connect experimental choices, observations, contemporaneous judgments and new-condition predictions. A separate thirty-session comparison tests access to research under matched initial scientific information and assessment. Research lowers average prediction error by 73% overall, yet increases it by 69% on three prespecified extreme-dilution queries. Records from the same comparison show a researcher revising an initial expectation, executing a new test and establishing a bounded local relationship, then giving that relationship substantial weight in an unresolved extrapolation. Corrected crystallization studies reveal a complementary divergence in the use of acquired evidence: forecasts outperform a source-observation mean for recovery in 39 of 42 sessions, but in none for purity. Autonomous research can thus establish useful local understanding while worsening some judgments beyond its tested range. Its contribution to scientific understanding depends on how experimental results are interpreted, how the scope of the resulting relationships is maintained, and how those relationships are used under new conditions.

An autonomous researcher does more than act on the world. By choosing experiments and measurements, it also shapes the experience from which its scientific judgments form: which phenomena become visible, which explanations encounter a test and which results guide the next investigation. The promise of autonomous science therefore concerns both the expansion of experimental capability and the transformation of experimental experience into scientific understanding.

Robot scientists and self-driving laboratories have demonstrated hypothesis testing, materials discovery and reaction optimization[3–6]. Language-model agents extend the connection between scientific instructions, tools and laboratory operations[7,8]. These advances make it possible to investigate a further question: how does an agent's own research change the relationships it uses to explain and predict the world? A successful procedure establishes an experimental achievement, and a correct forecast establishes predictive accuracy. Neither alone reveals the evidence from which a judgment arose or the scope that evidence supports[9,10].

We study this process in ChemWorld. Programmable world laws and independently specified public information make it possible to examine how initial knowledge enters experimental choices, how observations support or change judgments, and how these judgments guide further inquiry. Our evidence follows the organization of research through the formation of working relationships to their use under new interventions. Successful local tests, warranted revisions, unsupported interpretations of execution and response-specific transfer need not share a single failure mechanism. Together they reveal how the scientific consequences of research depend on the transitions between choosing evidence, interpreting it and applying what it supports.

Making autonomous scientific inquiry experimentally tractable

ChemWorld combines materials, operations and instruments into stateful virtual chemical systems. Agents choose quantities, processing sequences, measurements, repetitions and subsequent experiments within a fixed resource allowance. Their choices change samples and determine which observations become available. An evaluator independently controls world parameters and public information, records committed operations and obtains withheld reference outcomes (Fig. 1). The agent does not receive those outcomes during research or assessment.

This arrangement makes the agent's research process experimentally accessible. Knowledge tests and supplied-data tasks assess important scientific abilities[11,12], and interactive discovery environments already connect action to observation[13–16]. ChemWorld connects an agent's evidence selection to evolving public judgments and their later use in composable chemical processes. An attempted operation can be rejected before changing the sample; an observation can support a relationship locally without establishing it under every compatible intervention. Execution records and independently evaluated world responses allow these distinctions to be checked.

The broad study comprises 240 Sol campaigns across six chemical families and 60 additional equilibrium campaigns from four other model configurations. The current process analysis combines 45 equilibrium sessions with 45 crystallization sessions conducted under a corrected, consistent recovery definition. Each system covers five fixed worlds, three information conditions and three fresh model sessions per condition. These 90 sessions contain 80 complete research-and-assessment chains and 83 valid forecast stages; incomplete sessions remain in the denominator. Earlier crystallization sessions affected by

a recovery-description mismatch are retained separately, rather than pooled with their replacement cohort.

Contemporaneous notes connect publicly stated judgments to the observations available at that time. After research, agents seal a scientific account, predict outcomes under twelve new interventions without laboratory access, and reflect without receiving reference truth. Structured process analysis identifies 92 observable judgment–experiment–revision windows across 65 of the current 90 sessions (Extended Data Fig. 1). The other 25 sessions lack a complete window under this recording definition; this does not establish an absence of scientific learning. A separate contemporaneous comparison assigns 30 fresh sessions to research or no research under a common assessment protocol (Fig. 4). These evidence layers answer complementary questions and retain separate denominators.

The self-organization and adaptation of scientific inquiry

Research starts from goals, available information and a constrained action space. Descriptive comparisons across 150 retained campaigns show how these starting conditions enter the evidence history. In corrected crystallization, all fifteen aligned-information sessions start with solvent S2 and catalyst C1, whereas all fifteen sessions without an instance dossier start with S0 and C0. Under the specified mismatched dossier, thirteen start with S2 and fourteen with C1. These choices locate the beginning of inquiry; they do not by themselves demonstrate learning.

The resulting solvent coverage differs, but breadth alone is an incomplete account of scientific exploration. Historical electrochemical discovery and optimization campaigns cover similar numbers of solvents, and both cover all four at the larger budget. Full path summaries and initial-choice counts are retained in Supplementary Information. The more informative question is how an observation changes what the researcher tests next.

A single trajectory from the new opportunity comparison, here called equilibrium study A, makes this transition visible (Fig. 2). Under Opaque information, with no instance-specific archival relationship, the initial note predicts a weak-acid trend: increasing concentration should lower pH and dissociation fraction. The first six experiments instead yield dissociation fractions near 0.064–0.079. The next note withdraws support for the expected trend over the measured range and proposes that another accessible component might be the relevant input. Batches eight to twelve actually use that component, yet the response remains near a plateau. Its sealed account retains the local relationship and leaves the hidden mechanism unresolved. This is a substantive sequence of expectation, contrary observation, a new candidate explanation and an executed test.

Other studies continue a scan already planned at the outset, narrow a parameter interval or repeat a borderline recipe. All are forms of organizing inquiry, but only some constitute newly proposed tests in response to feedback. In the corrected crystallization cohort, 52 of 58 selected immediate plans match the ensuing committed actions and six match partly. This measures a public plan–action connection within observable windows; it is neither a discovery rate nor evidence that writing the note caused the action. The next question is what those actions and observations establish.

The formation and evolution of scientific understanding through experimentation

Scientific relationships acquire practical meaning when they support a further test, constrain an explanation or guide a new intervention (Fig. 3). In a corrected crystallization study, a near-threshold fines result leads to a prospective proposal to lower the reagent charge from 0.03 to 0.02 mol. The next batch implements that proposal and passes the fines threshold. The result supports a local recipe improvement. Because the absolute catalyst amount remains fixed, however, the catalyst-to-reagent ratio also changes: the experiment does not separately identify a concentration mechanism.

A second crystallization study tests an interpolation. After observing different recovery and fines outcomes at nearby solvent volumes, the researcher proposes that 0.065 L should preserve the recovery benefit while improving fines. The next experiment instead increases fines from 0.512 to 0.601, and the final note retracts the expectation. An unsuccessful operating intervention therefore produces an explicit, evidence-responsive revision. Together with the successful lower-charge test, it shows two ways experimentation can advance a working judgment: establishing a local improvement or withdrawing a prediction the observation does not support.

Whether an intervention actually occurred also determines what can be learned from it. In seed case B, a researcher plans approximately 1 g of cumulative seed addition, but only 0.05 g commits and nineteen subsequent requests are rejected. Its later note correctly recognizes that the high-dose test remains unexecuted. Seed case C encounters analogous rejected requests yet describes 1 g as tested and uses the outcome against a seed-based explanation. These contrasting accounts concern the evidential meaning assigned to the same kind of execution limitation. Additional thermal-history and near-threshold repetition cases remain in the supplementary record.

The same equilibrium study A now connects a warranted local revision to its later use (Fig. 2c,d). When forecasting an extremely dilute condition, the agent writes: “The point estimate still favors the experimentally supported fixed acidic background”. It predicts dissociation of 0.085 with an 80% interval of 0.035–0.350; the withheld reference mean is 0.716. Both its scientific account and forecast recognize that this regime was not tested. The transition is therefore more specific than an omitted caveat: a relationship supported locally receives substantial weight in a prediction whose continuation the experiments have not resolved. The matched no-research forecast for the same query is 0.569, with interval 0.040–0.980, and explicitly relies on an untested prior assumption. Its greater numerical accuracy does not establish a better experimental discovery.

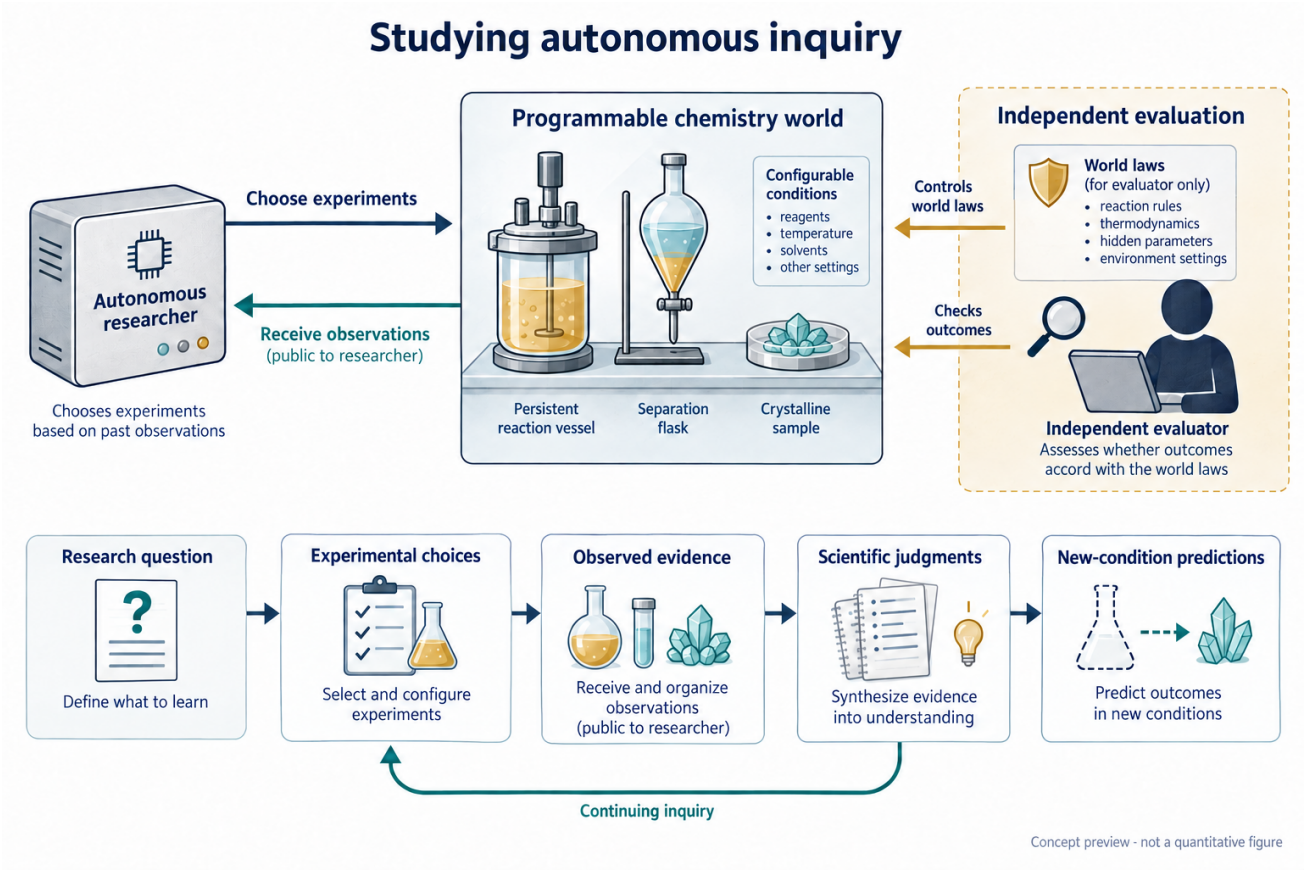


Figure 1 | Making inquiry experimentally tractable. Temporary ImageGen concept illustration. An autonomous researcher selects interventions and receives public observations from a persistent chemical world. The evaluator separately controls hidden parameters and checks world outcomes; reference outcomes are not returned to the researcher. The lower sequence links questions, choices, observations, public judgments and predictions, with judgments informing further choices. It summarizes observable relations, not a direct measurement of private cognition or a demonstrated causal effect of notebook text.

That continuation is not inevitable. Two sessions in another fixed equilibrium world both observe plateaux at relatively high loading, yet their dilute macro errors are 0.16287 and 0.01462. For the same dilute query, they predict dissociation of 0.130 and 0.626 against a reference mean of 0.65463. The more accurate session uses a weak-acid continuation consistent with its plateau observations and explicitly describes it as conjectural. Accurate extrapolation does not require direct sampling of every target condition; neither does accuracy establish that research experimentally identified the successful explanation. Continuous records distinguish supported local relations, unresolved possibilities and their subsequent predictive use.

The nature and conditions of scientific progress in autonomous inquiry

The process records locate how judgments form and are used; the matched research-opportunity comparison tests whether access to research changes subsequent prediction (Fig. 4). All thirty sessions use Opaque information: the public task, material identities, measurement scales, legal operations and initial state are available, but no instance-specific archival law or fitted relationship is supplied. Each of five equilibrium worlds receives three fresh sessions with twelve source batches and three without source experiments. Scientific inputs and sealed assessments are matched within each world and repeat. All thirty sessions terminate, all 180 planned source batches complete, and twenty-nine sessions provide valid forecasts. One research session loses its forecast and later stages to a network interruption; its observations and scientific account remain retained.

Research substantially improves prediction overall. Averaging valid sessions within each world and then weighting the five worlds equally, normalized macro mean absolute error falls from 0.132086 without research to 0.035632 with research, a 73.0% reduction. The direction holds in all five worlds. Yet for the three prospectively designated extreme-dilution queries, error rises from 0.061467 to 0.103677, a 68.7% increase, with deterioration in four of five worlds. Over the remaining nine queries it falls from 0.155625 to 0.012950. The comparison concerns the total opportunity to conduct research, including experiment selection, feedback and available recording tools; it does not isolate any one component.

The same research opportunity therefore improves overall prediction while worsening it in a defined region. All three response-specific point errors increase on the dilute subset, but by different amounts: Research-minus-NoResearch MAE is +0.00777 for normalized pH, +0.10795 for dissociation fraction and +0.01092 for precipitation signal. Dissociation accounts for 85.2% of the macro-error increase.

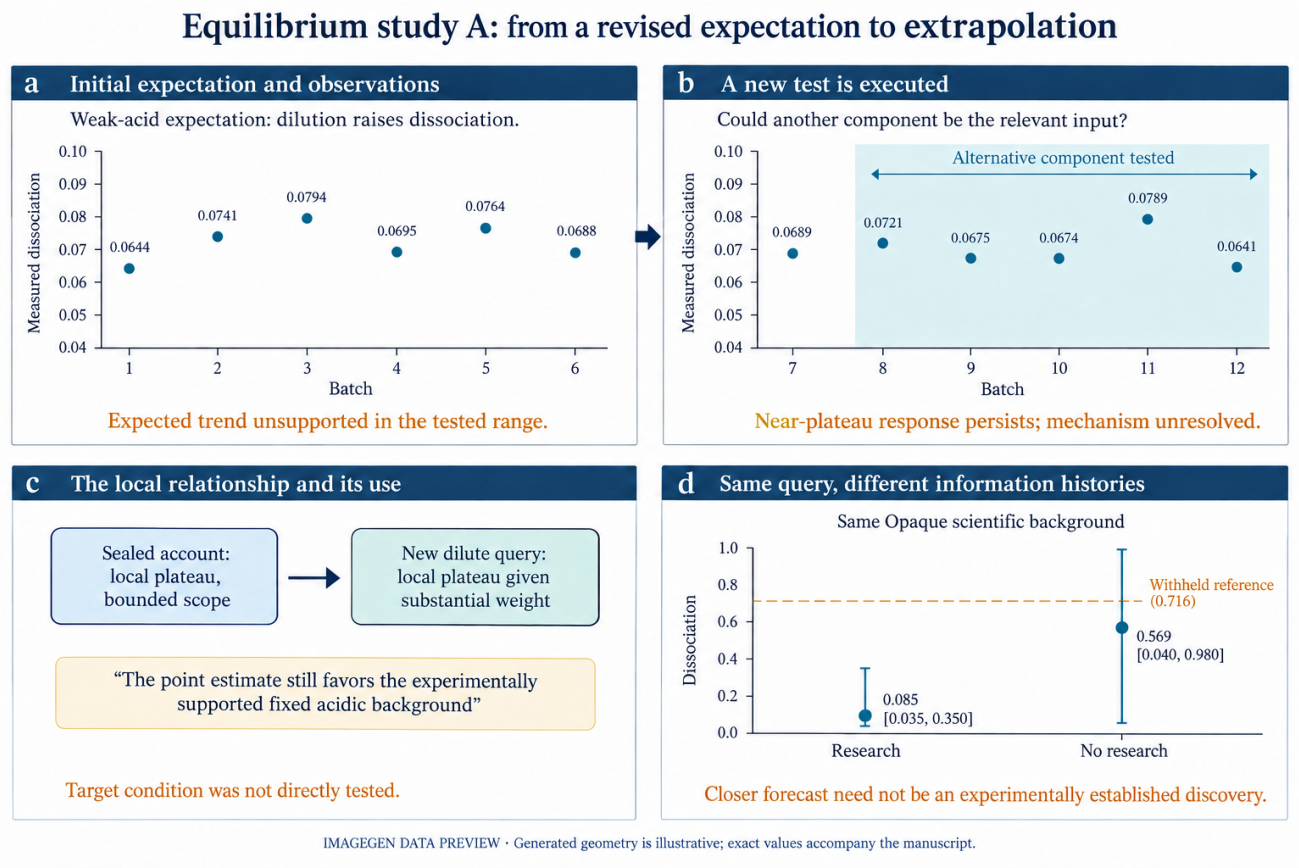


Figure 2 | Equilibrium study A, from a revised expectation to extrapolation. ImageGen data preview of one continuous Research session and its matched NoResearch forecast. a, The first six assays contradict the expected weak-acid trend within the tested range. b, A new candidate explanation motivates alternative-component tests in batches 8–12; the near-plateau persists. c, The bounded relationship in the sealed account is then given substantial weight in a new dilute prediction; the quotation is verbatim. d, Research predicts dissociation 0.085 [0.035, 0.350], NoResearch 0.569 [0.040, 0.980], against withheld mean 0.716. Intervals are agent-stated 80% predictive intervals; source points are individual assays. This is the new opportunity-comparison case, with identity preserved in the evidence archive. Generated geometry is illustrative.

This is an aggregate decomposition, not a claim that every query, world or session deteriorates. Figure 4 shows individual sessions and world means; the complete session-by-response and query-level results accompany the manuscript. Equilibrium study A supplies one observed route from a valid local finding to a distant judgment; it does not establish that this route explains every session-level difference.

The result extends beyond point error without yielding a universal story of misplaced confidence. At extreme dilution, research produces lower empirical coverage than no research for all three nominal 80% intervals, while the no-research intervals are substantially wider. An interval score penalizing both width and misses favours research for pH, favours no research for dissociation fraction and is similar for precipitation signal. Across all queries, research improves interval scores for all three responses. A conservative sensitivity analysis allowing the missing session's macro error to take any value in its legal range does not reverse either the overall benefit or the dilute deterioration (Methods).

Crystallization exposes a complementary pattern in the use of evidence already acquired (Extended Data Fig. 2). For each of 42 valid forecasts, transparent references use only that same session's source observations. Five-world-equal purity error is 0.03853 for the agent and 0.00510 for the overall source mean, which is more accurate in every valid session. Recovery shows the opposite pattern: agent error is 0.10129 against 0.20443, with the agent outperforming the mean in 39 of 42 sessions. Conditioning on the exact committed solvent set preserves both aggregate directions. Size and fines yield 5/42 and 23/42 wins, respectively. These references diagnose how acquired evidence is used across responses; they are neither no-research controls nor alternative autonomous researchers, and do not estimate the causal benefit of crystallization research.

These observations complement earlier operating–predictive comparisons. In electrochemistry, optimization produces a better recommendation in 26 of 30 matched conditions; among those 26, prediction is better in thirteen and worse in thirteen. In the repeated equilibrium cohort, aligned partial knowledge reduces error over nine queries from 0.00907 to 0.00512 while increasing dilute error from 0.09880 to 0.13205. Initial information, experimental achievement and subsequent prediction each contribute to the scientific outcome, but none substitutes for examining how research supports a relationship and how that relationship is used.

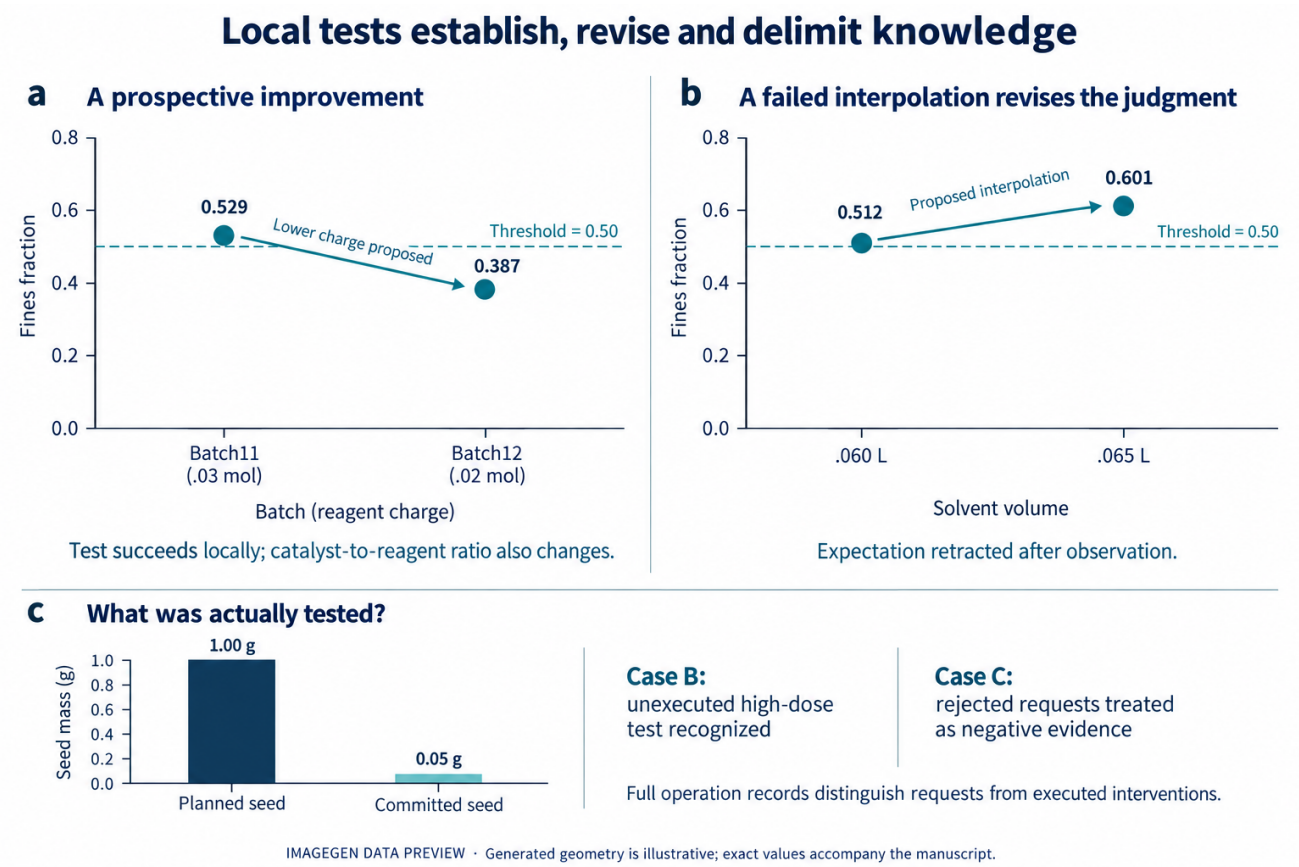


Figure 3 | Local tests establish, revise and delimit knowledge. ImageGen data preview of distinct crystallization studies. a, Lower charge reduces fines from 0.529 to 0.387, crossing the 0.50 threshold while also changing the catalyst-to-reagent ratio. b, A solvent-volume interpolation raises fines from 0.512 to 0.601; the researcher retracts its expectation. Points in a and b are individual assays, not averages with uncertainty intervals. c, Planned 1 g seed additions leave only 0.05 g committed. Case B recognizes an unexecuted high-dose test; case C treats rejected requests as negative evidence. These are separate researchers, not successive states of one trajectory. Generated geometry is illustrative; complete records accompany the manuscript.

Discussion

Autonomous research can produce useful local understanding without ensuring that its subsequent application remains within the scope supported by evidence. In the matched comparison, research substantially improves prediction overall while worsening predictions in a prespecified dilute region. A continuous trajectory shows that this divergence can coexist with appropriate revision and new experimentation: the researcher rejects an initial expectation, tests a competing possibility and forms a bounded empirical relationship, then weights that relationship heavily when judging a condition it has not resolved. The crystallization references expose a complementary limitation: an acquired research history can support effective prediction for one response while another already-observed relationship is poorly retained. These are concrete differences in what research contributes and how its results are used.

Studying the formation of scientific understanding requires specifying which relationship is proposed, what information supports it at that point, what the world actually tests and how the relationship is later used. The records include warranted retractions and bounded improvements alongside interpretations unsupported by committed operations. They also include accurate forecasts grounded in possibilities that remain experimentally unresolved. These distinctions are consistent with the role of discriminating tests and inference in scientific inquiry[17–19], but their implementation by autonomous agents cannot be assumed from successful tool use or a plausible scientific account.

The controls delimit what can be explained causally. The matched opportunity comparison estimates the effect of the full research package under a common assessment protocol. A separate fixed-recipe quench intervention identifies physical effects of an operation in the virtual worlds, without providing outcomes to an adapting agent (Extended Data Fig. 3). Neither identifies a notebook's independent effect or proves that a particular judgment maintains itself through selective experimentation. Explaining the observed formation and use of relationships does not require every trajectory to share such a mechanism or end in a successful corrective strategy.

The study is limited to selected programmable systems, a small fixed set of worlds and, for repeated process analyses, one model configuration. Fresh sessions are not controlled model-seed replications or independent samples of world laws. Notebook statements are public records rather than access to private cognition; gaps restrict temporal inference. Cases are retrospectively selected from systematically retained records, with single-analyst interpretation

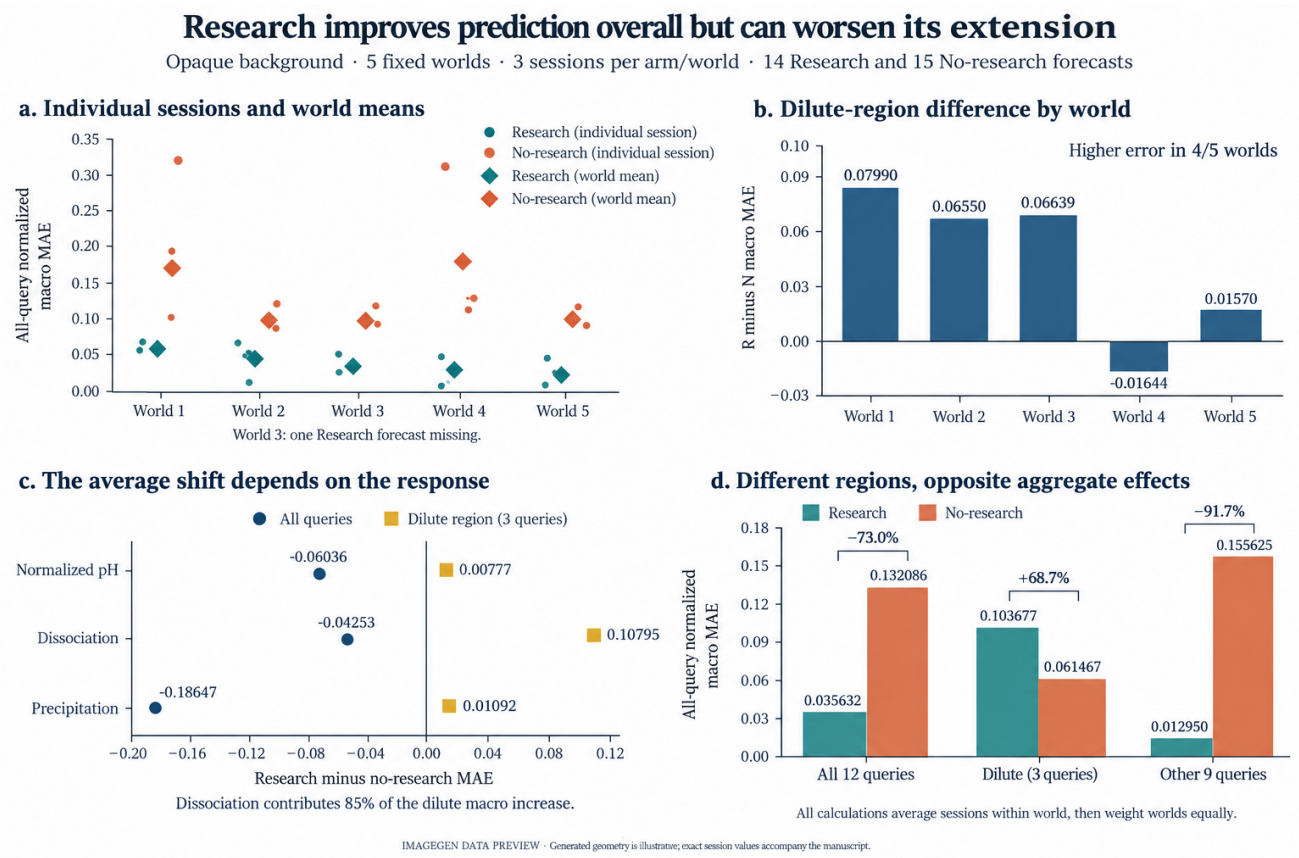


Figure 4 | Overall gains and a regional reversal from research. ImageGen data preview. All thirty sessions use Opaque information and matched scientific inputs. a, Individual all-query macro errors and within-world means; Research has fourteen valid forecasts and NoResearch fifteen. World 3 has one retained missing Research forecast. b, Dilute-region Research-minus-NoResearch macro error is positive in four of five worlds. c, Response-specific shifts differ; dissociation contributes 85.2% of the dilute macro increase. d, Equal-world errors across all twelve, dilute three and other nine queries change by −73.0%, +68.7% and −91.7%, respectively. Generated dots and geometry are illustrative; exact session values, response errors and all 1,044 valid query-response rows are supplied in the evidence archive. No confidence intervals or population sampling inference are implied.

and no second-rater reliability estimate. Historical and current protocols retain separate provenance, including the corrected crystallization block and all failures. These limits constrain claims about prevalence and generality while leaving the observed transitions and bounded comparisons available for scrutiny.

The value of autonomous research therefore depends on more than the accumulation of valid experimental results. It also depends on which relationships those results establish and how the researcher carries their evidential scope into subsequent inquiry. The present experiments show substantial gains from research alongside a reversal of benefit in a defined prediction region; their records connect this outcome to observable transitions from expectations to local findings and further judgments. Scientific progress in autonomous inquiry is produced through the continuing interpretation and use of evidence. ChemWorld makes those transitions available for experimental comparison and scrutiny.

Funding

This work was supported by the Shenzhen Science and Technology Program (AI2026004), the Tsinghua University Initiative Scientific Research Program and the Scientific Research Innovation Capability Support Project for Young Faculty (ZYGXQNJSKYCXNLZCXM-E7).

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Methods

Worlds, interfaces and experimental units

ChemWorld compiles compatible process components into persistent virtual chemical environments. The public contract specifies materials, legal operations, instruments, resources and task objectives. Private parameters determine each world's responses. Agents access the laboratory and public records, without evaluator state, simulator code, private parameters or withheld reference outcomes. Operations change samples; measurements may consume material. Candidate transitions are validated before commitment. Rejected operations do not constitute the requested physical intervention, although attempts can consume contract-defined allowances. Exact replay checks environment transitions, not regeneration of model decisions.

The broad study covers electrochemistry, partitioning, reaction and thermal processing, aqueous equilibrium, reaction-coupled crystallization and purification. These are controlled virtual systems, not experimentally validated predictions for named physical substances. Current repeated studies use equilibrium and crystallization. Five fixed worlds recur across arms and repetitions; their parameters, initial states and keyed observation mappings are fixed. Repetitions are fresh model sessions, not independent world draws or controlled model-seed replications. The model interface does not expose a usable sampling seed. Repeated identical interventions can share keyed observations, so they are not independent noise samples.

Evidence layers and retained denominators

The historical broad pool contains 240 Sol-medium campaigns and 60 additional equilibrium campaigns from Luna, Terra, GPT-5.5 and Astra, requested at medium reasoning effort. The five-model equilibrium comparison includes fifteen Sol campaigns already in the 240. Requested model labels do not establish equal realized computation. The original 240 include 60 electrochemical, 30 partitioning, 60 reaction, 45 equilibrium, 30 crystallization and 15 purification campaigns. They retain 3,597/3,600 final assays, all 720 assessment stages and 238 protocol-conforming campaigns under their original rules. The additional sixty retain 720 batches and 240 stages. Two original source-completion failures remain included. Earlier selected notebook studies comprise another 24 sessions and are not pooled into the broad or current process cohort.

The current process cohort combines the 45 equilibrium sessions from the earlier 48-session block and its 42-session extension with a complete replacement block of 45 crystallization sessions under a consistent recovery description. Each system covers five worlds, Opaque/Aligned/MisIndexed information and three sessions per cell, with twelve research batches per session. Equilibrium retains 39 complete chains and 41 valid forecasts; corrected crystallization retains 41 and 42, respectively. The combined current denominator is 90 sessions, 80 complete chains and 83 valid forecasts. Old crystallization sessions remain protocol-affected historical evidence. They are not added to the replacement sessions when calculating current outcomes or process frequencies.

Corrected crystallization has 510/540 source batches, 125/135 assessment stages, 145 notebook versions and 42 recommendation retests. All 45 source trajectories and 42 retests pass replay. Four incomplete chains remain: three lack forecasts, and one retains a valid forecast but lacks the subsequent retrospective. The equilibrium subset and the separate opportunity block retain their own failure tables. Missing outcomes are not set to zero, and later favourable outcomes do not replace completed results. Evidence integration does not change the original development or qualification status of any block.

Information, autonomy and recording

Opaque supplies the public task and operational context without the additional instance dossier. Aligned supplies partial information appropriate to the system. In equilibrium, this includes an effective-acidity interval and an anchored local dilution relationship. MisIndexed uses another world's archival relationship with the same format and qualification. In crystallization, Aligned gives nominal descriptors for anonymous materials; MisIndexed swaps solvent S1/S3 descriptors while preserving S0, S2 and catalyst descriptors. Neither dossier specifies the complete world law.

The original equilibrium Opaque/Aligned sessions precede its MisIndexed extension. Corrected crystallization is a later complete block. These descriptive comparisons are not all concurrently randomized treatments, and new versus old crystallization is not an estimate of the causal effect of correcting the prompt. The new opportunity comparison uses a common contemporaneous protocol, described below.

Agents choose operations, quantities, measurements, repetitions and grouping within fixed budgets. Equilibrium permits twelve source batches, 180 operation attempts, twelve intermediate instruments and twelve final assays; crystallization uses its fixed twelve-batch materials envelope and 720-attempt ceiling. A notebook role asks for an initial testable expectation when available; subsequent updates are agent-selected, without a write-every-batch requirement or automatic reinjection. The role may influence research and is not assumed passive. Notes retain version and public-event cursor. These public records do not expose private cognition. Absence of a notebook-read tool call does not establish absence of access to the conversation's earlier content.

Assessments and recovery definition

After source research and any operating recommendation, agents seal a scientific account, answer twelve fixed new-intervention queries without laboratory tools and then reflect without truth feedback. Equilibrium includes a structured supplemental assessment. The evidence notation K1/Q/K2/EQS denotes these stages. Queries are hidden during research; post-assessment proposals are not treated as executed source experiments.

Equilibrium errors compare predictions with the mean of five reference observations per query. Coverage checks all five observations. Crystallization uses the noiseless full-process response immediately before final-assay sampling; source observations are noisy. The corrected block retains sixty reference batches, numerically matching the earlier reference outcomes, and separately retests each available committed recommendation.

The corrected recovery definition is seed-excluded recovered solid target divided by the tracked initial reagent charge, in moles, clipped to [0,1]. Source instructions, prediction prompts, observations and scores now refer to this same full-process quantity. The internal crystallization-stage metric instead divides by target product before separation. Earlier process instructions described this latter denominator while observations and scoring used initial reagent charge. Original scores are preserved as protocol-affected diagnostics; they do not enter the corrected cohort. The new block was rerun in full rather than retrospectively relabelling those scores.

Quality requires particles, purity at least 0.80 and fines at most 0.50; joint recommendation success also requires corrected recovery at least 0.10. Corrected joint counts are 13/15 Opaque, 10/14 Aligned and 10/13 MisIndexed among available retests, against fifteen planned sessions in each arm. Particle size is the public index min(d50/250 micrometres, 1); its error is not measured in micrometres.

Research paths and process interpretation

Retrospective path extraction covers sixty historical electrochemical, forty-five equilibrium and forty-five corrected crystallization sessions. Fifty within-world condition contrasts retain goal, budget, information and cohort. Counts use committed operations; solvent coverage includes additions during resuspension. Rejected attempts do not establish successful material use. Interrupted sessions remain in the planned denominator.

Structured extraction considers all 178 adjacent pairs among 268 notebook versions in the current 90 sessions. Eligibility requires an earlier note, at least one new committed final assay before the next note, and a committed research operation after that later note; measurement and termination alone do not satisfy continuation. All 178 pairs contain new assays. Ninety-two have the required continuation: 34 equilibrium and 58 crystallization windows in 30 and 35 sessions, respectively. The other 86 pairs (44 equilibrium, 42 crystallization) have no later committed research action and remain available in the evidence archive. The 25 sessions without a complete window are not classified as having learned nothing.

Eligibility is computed from event timing before semantic plan matching, without testing whether a proposed action succeeds. Nevertheless, the rule conditions the sample on continued committed research and excludes terminal revisions from this particular action-linked denominator. All 92 eligible windows are coded. Labels concern one selected immediate plan, not every proposal in a note: 83 matched, seven partly matched and two without an explicit next plan; corrected crystallization contributes 52 matched and six partly matched. Full notes retain abandoned, rejected and altered proposals even when another selected action matches. All 182 update/plan quotations and their assay/action cursors are verified against source records. The archive includes every adjacent pair, selected code, full note and execution history.

The same analyst performs coding and internal scientific review. There is no independent second-rater agreement estimate, blinding or claim of adjudicated mechanism prevalence. The team has assigned review to this workflow; completed internal review is distinguished from independent replication. Counterexamples include warranted retractions, correct recognition of unexecuted tests and accurate but explicitly conjectural extrapolation. Selection of illustrative cases is retrospective.

Source-only forecast references

Two transparent references are computed for each valid corrected-crystallization forecast using only that researcher's successful source final assays. The first averages all available assays. The second averages assays whose exact set of committed solvent identifiers matches the query, falling back to the overall mean when none match. Solvent sets include successful solvent additions and resuspension, excluding rejected actions. The fitting function receives no test truth. These references are descriptive calculations, not separately trained agents or mechanistically complete baselines.

There are 42 valid sessions and 504 queries. Same-solvent fallback occurs in 186/504 queries: 1/180 Opaque, 95/168 Aligned and 90/156 MisIndexed. Matching does not control thermal history, quantities or final liquid composition. References have no predictive intervals. Each response is reported in its own public scale; no combined score is formed across recovery, purity, size and fines.

Matched research-opportunity comparison

The fixed design contains five equilibrium worlds, Research/NoResearch and three fresh Sol-medium sessions per cell, for thirty planned sessions. All use Opaque information. The shared public background contains the task and measurement definitions, material catalogue, legal operations/instruments and initial state; the instance archival prior is null. Opaque therefore means absence of an additional instance-specific law, not absence of public chemical framing or model pretraining. The complete initial public contexts accompany the manuscript. Fifteen within-world/repeat checks confirm scientific-input equality after removing only the non-scientific campaign identifier. Research additionally receives twelve autonomous source batches and the existing notebook opportunity; NoResearch receives no source operations or observations. Both arms use identical assessment prompts, schemas, order and restrictions.

All thirty sessions reach terminal status. Research completes 180/180 batches; the combined arms complete 117/120 assessment stages and provide 29 valid forecasts. In world three's third Research repetition, the forecast stream disconnects after the scientific account. Q and the two later stages are missing, without replacement. Thus Research has fourteen valid forecasts, NoResearch fifteen; Research world three has two rather than three. Fifteen source trajectories and twenty-five reference trajectories (300 reference batches) pass exact replay. The retained resource ledger is complete for 29/30 sessions; one has only recorded partial usage. Recorded input plus output totals 13,428,656 tokens. Missing usage is not zero and monetary cost is unavailable.

A first execution attempt failed because the assessment transport configuration omitted a required field. Four launched sessions, two completed source histories totalling twenty-four batches and their resource costs are retained separately; no scientific-account stage completed. After fixing that execution defect, the whole fixed thirty-session block restarted. This was a protocol repair, not replacement based on scientific outcomes. Only the complete restarted block enters the comparison.

The comparison estimates the total research-opportunity package under a shared assessment. It does not isolate autonomy relative to supplied data, feedback alone, notebook use or private representational change. It is between fresh sessions, without a within-session pretest. World and arm order are balanced in a predetermined launch schedule, not a random sample of world laws.

Quantitative summaries and missingness

Equilibrium macro MAE equally weights normalized pH, dissociation fraction and precipitation signal across the designated queries. The extreme-dilution group comprises three queries fixed before the repeated blocks; the remaining nine are not assumed entirely in distribution. Session errors are averaged within each world, then the five worlds receive equal weight. Individual sessions and all missing stages remain visible. Responses, queries, reference observations and batches are correlated within sessions and worlds; they are not independent researcher samples.

Coverage and mean interval width accompany the proper interval score for the agents' nominal 80% intervals[20]. With lower endpoint l, upper endpoint u, outcome y and alpha = 0.2, the score is (u-l) plus (2/alpha)(l-y) when y is below l, or (2/alpha)(y-u) when y is above u. Lower is better. Aggregates weight worlds equally. Agent intervals are not confidence intervals for our group means.

A post hoc sensitivity analysis assigns the single missing opportunity forecast any macro MAE in [0,1], the legal error range, and recomputes means using the planned three sessions per arm/world and five equal worlds. Research-minus-NoResearch bounds are [-0.098659, -0.031992] overall and [+0.036768, +0.103434] for dilution. These are conservative bounds on missing-outcome influence, not imputation, confidence intervals or population inference. Three repetitions per fixed condition describe variation without establishing a population reliability rate.

Matched quench intervention

A separate retained block applies the eighteen original first-cohort crystallization recommendations in five worlds, with and without a single quench operation: ninety pairs, 180 trajectories. All other action arguments and initial states are matched. Quench is removed at its recorded location or inserted immediately before the first seed/crystallization-cooling operation. No agent adapts or receives these outcomes.

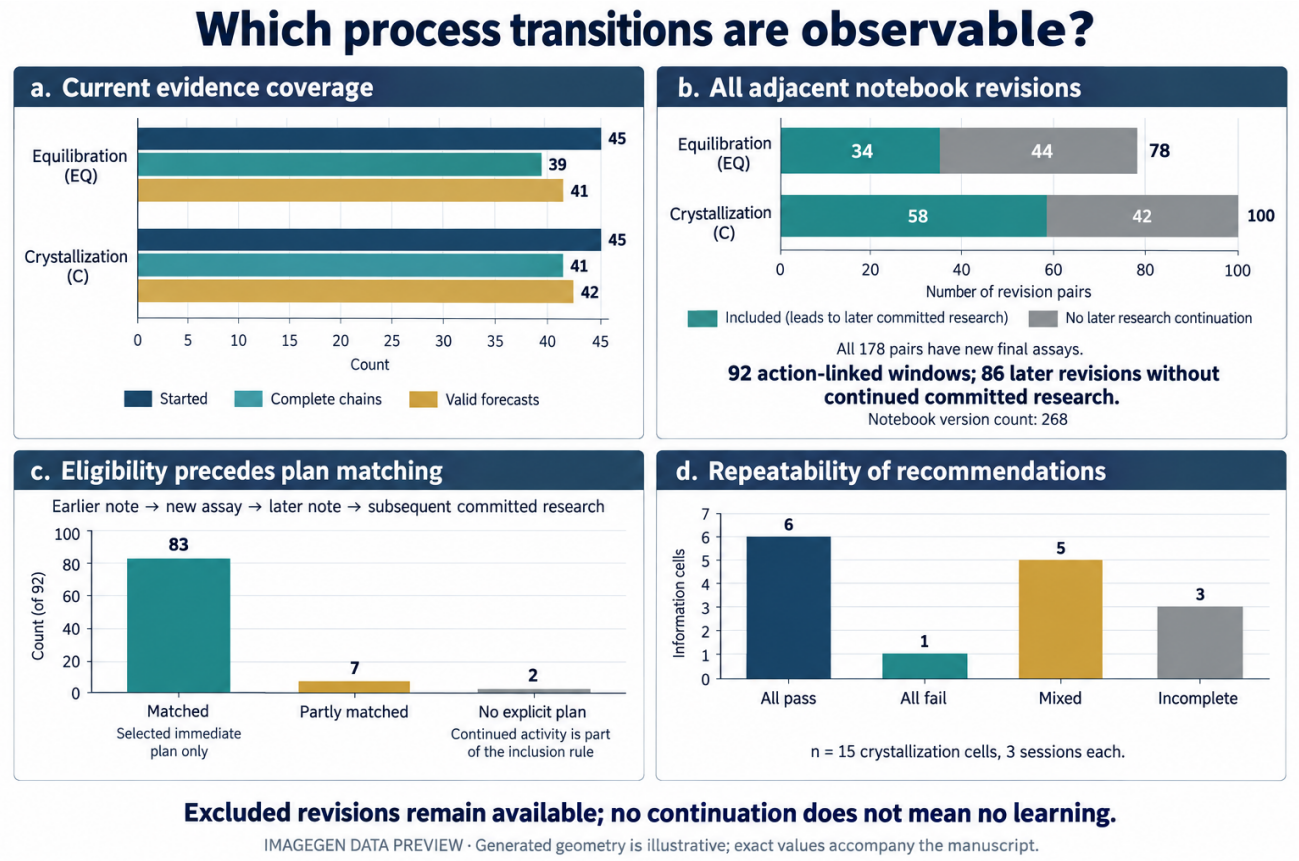
All trajectories are retained and replayed; seventy-five pairs reach both final assays. In five, the absent arm lacks particles; in ten, the present arm makes the subsequent fixed cooling request invalid. Failed arms are not repaired. Among complete pairs, mean present-minus-absent fines changes by +0.30937 and recovery by +0.00741; joint success changes from 34 to 4. These conditional effects describe physical operations under the implemented initial-charge recovery metric. Quench changes multiple physical state variables and does not isolate cooling rate or a cause of scientific judgment. Donor recipes are historical and are not sampled from the corrected block.

Data and code availability

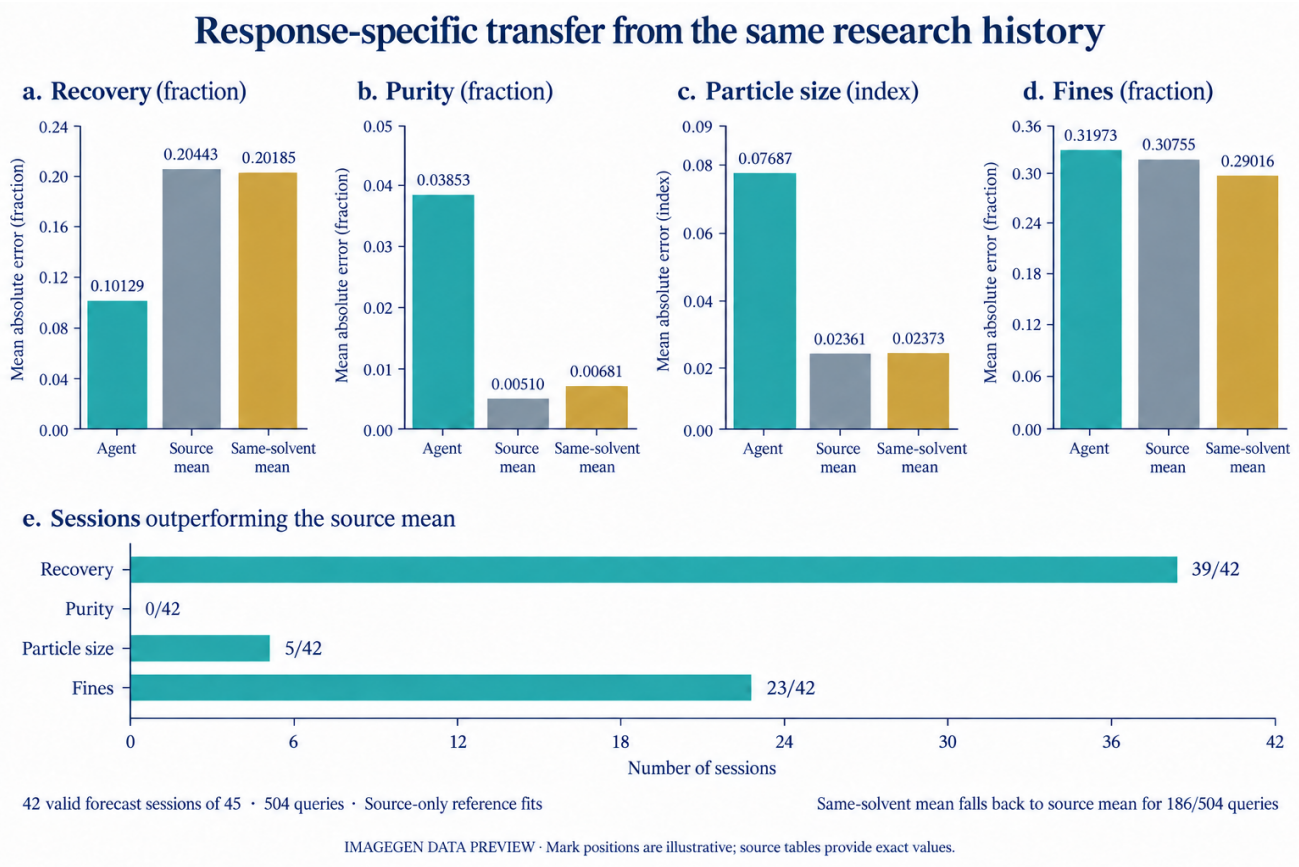
A portable review archive accompanies the manuscript with complete public initial inputs, 1,044 query-response prediction rows, all 270 planned session/group/response cells, world summaries, the continuous focal study, all 178 adjacent note pairs and full public histories for the current 90 process sessions. Missing forecasts remain explicit. Original summaries, exact replay and resource records remain in the project repository; the archive adds no participant evidence or independent second-rater assessment. Supplementary Information reports the numerical decomposition and coding-selection rules. A persistent public deposit remains to be finalized. Manuscript compilation reads retained results without new model or simulator experiments.

Figures are temporary ImageGen previews: one framework illustration and six data-layout previews. Numerical anchors come from retained evidence; generated marks remain illustrative. Exact results are reported in the text and tables. Prompts and provenance accompany the artwork.

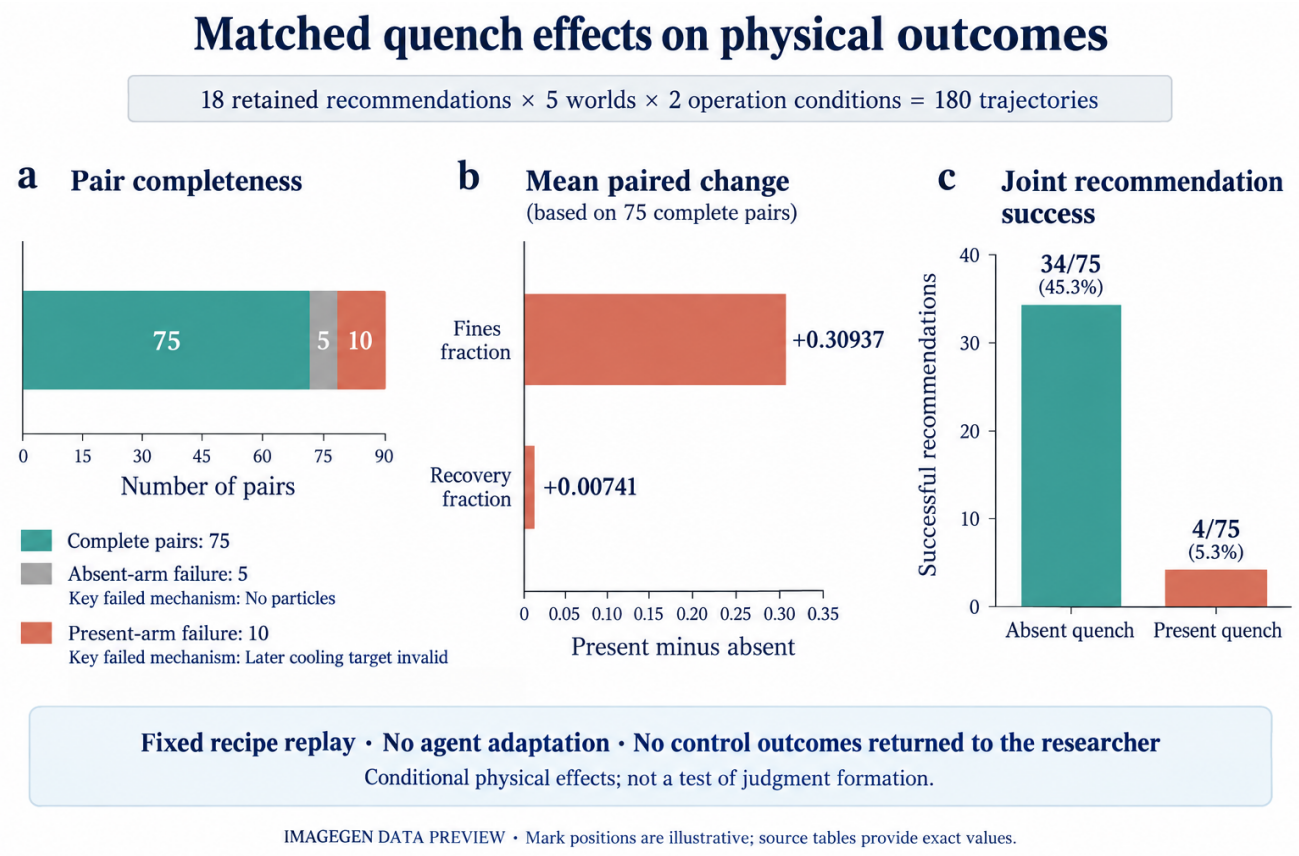
Extended Data



Extended Data Figure 1 | Which process transitions are observable? ImageGen data preview. a, Current session coverage and valid forecasts. b, All 178 adjacent version pairs contain new assays: 34/78 equilibrium and 58/100 crystallization pairs also have a later committed research action, yielding 92 eligible windows. The remaining 86 revisions are retained without an action-linked code. c, The selected immediate plan is matched in 83 windows, partly matched in seven and absent in two. Eligibility precedes semantic matching but conditions the sample on continued activity; these counts do not measure learning success or cover every proposal. d, Across fifteen crystallization world-by-information cells with three sessions each, six are all-pass, one all-fail, five mixed and three incomplete. Generated geometry is illustrative.



Extended Data Figure 2 | Response-specific transfer from the same research history. ImageGen data preview. a–d, Agent forecasts and two source-only references are compared for recovery, purity, particle-size index and fines, retaining each response's own public scale. Particle size is the dimensionless index min(d50/250 micrometres, 1). e, Agent forecasts outperform the overall source mean in 39/42, 0/42, 5/42 and 23/42 sessions, respectively. The 42 valid corrected-crystallization sessions supply 504 forecast queries. Same-solvent means fall back to the overall mean in 186/504 queries and do not match thermal history or quantities. Neither reference is fitted to held-out truth. Generated geometry is illustrative; five-world-equal numerical errors are reported in the supplementary tables.



Extended Data Figure 3 | Matched quench effects on physical outcomes. ImageGen data preview. Eighteen historical recommendations are replayed in five worlds with and without one quench operation, yielding ninety pairs and 180 trajectories. a, Seventy-five pairs yield both final assays; five absent arms lack particles and ten present arms invalidate a subsequent fixed cooling target. b, Complete-pair mean changes are +0.30937 in fines and +0.00741 in recovery. c, Joint success changes from 34/75 without quench to 4/75 with quench. These are conditional physical effects under the initial-charge recovery definition. The agent neither adapts nor receives control outcomes, so this intervention does not identify judgment formation. Generated geometry is illustrative and all failed pairs remain reported.